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Truncated Tetrahedron

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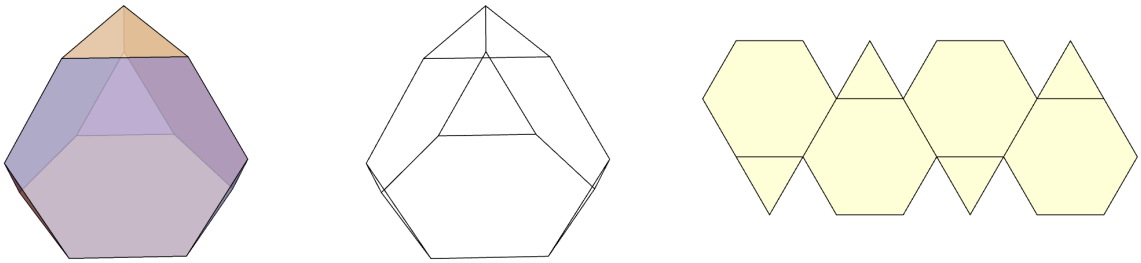

MAKE YOUR OWN TRUNCATED TETRAHEDRON
Print & fold
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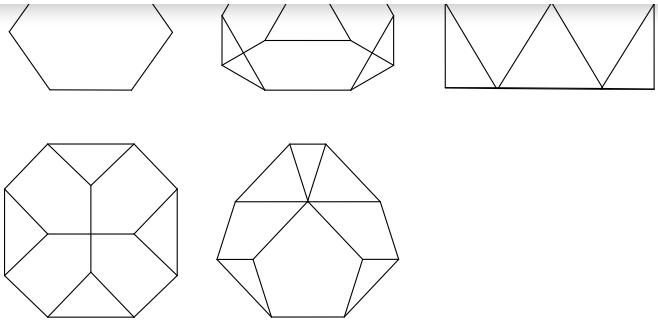
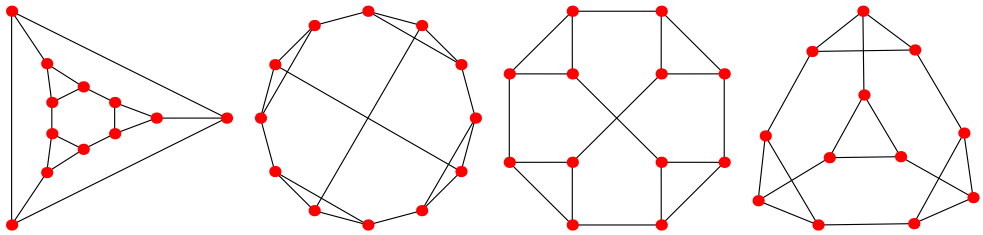
The truncated tetrahedron is the Archimedean solid with faces $4\{3\}+4\{6\}$. It is also the uniform polyhedron with Maeder index 2 (Maeder 1997), Wenninger index 6 (Wenninger 1989), Coxeter index 16 (Coxeter *et al.* 1954), and Har'El index 7 (Har'El 1993). It has Schläfli symbol $t\{3,3\}$ and Wythoff symbol $23\mid 3$. It is illustrated above together with a wireframe version and a net that can be used for its construction.

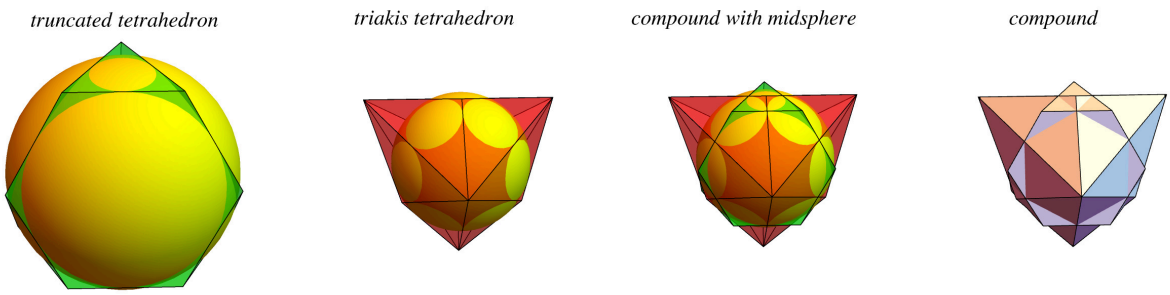


Some symmetric projections of the truncated tetrahedron are illustrated above.

It is implemented in the Wolfram Language as PolyhedronData["TruncatedTetrahedron"] or UniformPolyhedron["TruncatedTetrahedron"]. Precomputed properties are available as PolyhedronData["TruncatedTetrahedron"].



The skeleton of the truncated tetrahedron is the truncated tetrahedral graph, illustrated above in a number of embeddings.



The dual polyhedron of the truncated tetrahedron is the triakis tetrahedron, both of which are illustrated above together with their common midsphere. The inradius r of the dual, midradius ρ of the solid and dual, and circumradius R of the solid for $a = 1$ are

$$r = \frac{9}{44}\sqrt{22} \approx 0.95940 \tag{1}$$

$$\rho = \frac{3}{4}\sqrt{2} \approx 1.06066 \tag{2}$$

$$R = \frac{1}{4}\sqrt{22} \approx 1.17260. \tag{3}$$

$$r_3 = \frac{5}{12}\sqrt{6} \tag{4}$$

$$r_6 = \frac{1}{4}\sqrt{6}\,. \tag{5}$$

The surface area and volume are

$$S = 7\sqrt{3} \tag{6}$$

$$V = \frac{23}{12}\sqrt{2}\,. \tag{7}$$

The unit truncated tetrahedron has Dehn invariant

$$D = 12\,\langle 3\rangle_2 \tag{8}$$

$$= 12\tan^{-1}\left(\sqrt{2}\right), \tag{9}$$

where the first expression uses the basis of Conway *et al.* (1999).

SEE ALSO

Archimedean Solid, Equilateral Zonohedron, Jabulani Polyhedron, Triakis Tetrahedron, Truncated Tetrahedron-Triakis Tetrahedron Compound

EXPLORE WITH WOLFRAM|ALPHA

WolframAlpha truncated tetrahedron

More things to try: = truncated tetrahedron = adjugate {{8,7,7},{6,9,2},{-6,9,-2}} = domain of f(x,y) = log(1-(x^2+y^2))

REFERENCES

Conway, J. H.; Radin, C.; and Sadun, L. "On Angles Whose Squared Trigonometric Functions Are Rational." *Discr. Computat. Geom.* **22**, 321-332, 1999.

Coxeter, H. S. M.; Longuet-Higgins, M. S.; and Miller, J. C. P. "Uniform Polyhedra." *Phil. Trans. Roy. Soc. London Ser. A* **246**, 401-450, 1954.

Cundy, H. and Rollett, A. "Truncated Tetrahedron. 3.6^2." §3.7.1 in *Mathematical Models, 3rd ed.* Stradbroke, England: Tarquin Pub., p. 101, 1989.

Geometry Technologies. "Truncated Tetrahedron." http://www.scienceu.com/geometry/facts/solids/tr_tetra.html.

Har'El, Z. "Uniform Solution for Uniform Polyhedra." *Geometriae Dedicata* **47**, 57-110, 1993.

Kasahara, K. "Three More Semiregular Polyhedrons Become Possible." *Origami Omnibus: Paper-Folding for Everyone.* Tokyo: Japan Publications, p. 225, 1988.

Maeder, R. E. "02: Truncated Tetrahedron." 1997. https://www.mathconsult.ch/static/unipoly/02.html.

Wenninger, M. J. "The Truncated Tetrahedron." Model 6 in *Polyhedron Models.* Cambridge, England: Cambridge University Press, p. 20, 1989.

Weisstein, Eric W. "Truncated Tetrahedron." From *MathWorld*--A Wolfram Web Resource.
https://mathworld.wolfram.com/TruncatedTetrahedron.html

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